Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/14/2005
PI Name:	Moorman, Stephen J Ph.D.		
Project Title:	The Effects Of Microgravity On Gene Expression In Live Zebrafish Embryos		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	08854-5635	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1591		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effects of microgravity on gene expression are poorly understood. Initial attempts have been made to investigate the effects of microgravity on cells in culture. It is unknown how those experiments will relate to potential effects on gene expression in vivo. We have begun to examine the effect of simulated-microgravity on gene expression in specific developing organ systems in vivo using transgenic zebrafish that express the green fluorescent protein gene (gfp) under the influence of different promoters. The use of gfp as a "reporter-gene" has two significant advantages; you can monitor gene expression directly in a live animal, and you can easily detect changes in morphology that might be due to changes in expression of other genes. This is the first time a whole animal in vivo model will be used and maintained alive during gene expression analysis for microgravity related research. This research not only addresses an important issue in space life sciences research, it clearly demonstrates the usefulness of zebrafish with gfp constructs as a model to		

Task Description:	study the effects of microgravity on gene expression during development. Use transgenic zebrafish that express the gene for GFP under the influence of specific promoters to determine the effects of simulated-microgravity on gene expression in live zebrafish embryos. We have documented specific changes in gene expression that are related to changes in gravitational force. This suggests that gravitational force plays a fundamental role in regulating gene expression and might have been a driving force for vertebrate evolution. Interestingly, organs that show effects in our experiments either develop into or regulate the development of organ systems that are affected in astronauts during space flight. This suggests that ground-based research using zebrafish embryos might have predictive value for the affects of long-duration space flight on astronaut health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth's gravitational field has remained constant during all of evolutionary history. Therefore, it is possible that gravity might play some fundamental role in regulating gene expression during vertebrate development. Interestingly, organs that show effects on gene expression in our experiments either develop into or regulate the development of organ systems that are affected in astronauts during space flight. This suggests that ground-based research using zebrafish embryos might have predictive value for the affects of long-duration space flight on astronaut health.
Task Progress:	We have completed our studies using the beta-actin:gfp fish, the hsp70:gfp fish, and the fli1:gfp fish. We have identified periods of susceptibility to the effects of simulated-microgravity for the major organ systems in the zebrafish embryo except for the nervous system. We have also identified recovery periods during which continued exposure to simulated-microgravity no longer affects the expression of our target genes. We are continuing to look at the effects of simulated microgravity on the developing nervous system and on the developing lens of the eye. We are focusing on the lens because it appears to be one of the most susceptible organs in the developing zebrafish.
Bibliography Type:	Description: (Last Updated: 07/05/2019)
Abstracts for Journals and Proceedings	Shimada, N.; Moorman, S.J. "Effects of microgravity on hsp-70 gene expression in developing lens." Experimental Biology 2005 Meeting FASEB J 491.9 , Apr-2005
Articles in Peer-reviewed Journals	Shimada, N; Sokunbi, B.; Moorman, S.J. " Changes in gravitational force affect gene expression in developing organ systems at different developmental times." BMC Developmental Biology 2005, 5:10. , May-2005